

Habituation-Aware Task Valuation for Multi-Robot Bird Deterrence Using Signal Temporal Logic

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Abstract—Repeated interventions can reduce the effectiveness of future actions, making task value depend on intervention history as well as current demand. Multi-robot bird deterrence exhibits this coupling: repeated cues lose effectiveness through habituation, whereas unused cues recover. We develop a history-dependent predictive task value using Signal Temporal Logic (STL). A local specification combines exposure, service recency, and cue-effectiveness requirements; candidate actions are ranked by their counterfactual improvement in a smooth aggregate of specification robustness. The value integrates into an existing decentralized dispatcher without modifying its task-admission or reservation mechanisms, and uses only cue-effectiveness states that each robot reconstructs from intervention history. In paired vineyard simulations with spare service capacity, the proposed system reduces value-weighted exposure by 9.3% relative to fixed-cue greedy dispatch and outperforms habituation-aware greedy dispatch by a further 3.9 percentage points. A two-factor comparison shows that the gain arises from combining the effectiveness requirement with flexible cue selection; the combination improves on either component alone by 10.4 percentage points. Non-habituating controls and load sweeps delineate the operating envelope: the benefit requires active habituation and spare capacity, reverses under sustained high demand, and trades against reactive deadline service, a trade-off we quantify and show to persist across dispatcher settings. These results establish the value and operating regime of specification-based task valuation for interventions with history-dependent effectiveness.

I. INTRODUCTION

Persistent intervention tasks differ from one-time service tasks because executing an action can change the value of later actions. In bird deterrence, a cue that is effective on its first application may become less effective after repetition. A robot must therefore decide not only where and when to intervene, but also which cue to use and which cues to allow to recover. This creates a coupling between immediate suppression and the effectiveness available for subsequent interventions.

Response decrement under repeated stimulation and recovery during rest are established features of habituation [1], which can limit acoustic deterrents [2]. Robotic deterrence offers mobility and control over intervention timing, and field studies demonstrate that robotic predators can disperse birds [3]. Response nevertheless depends on the stimulus and species: the RobotFalcon study found no evidence of habituation during its observation period. We consequently treat habituation as a scenario-dependent mechanism, rather than an inevitable property of every deterrent. A useful

allocation policy should exploit this mechanism when it matters, and its cost should be evaluated when it does not.

The technical question is how to assign predictive task value when intervention effectiveness is itself a dynamic state. We use Signal Temporal Logic (STL) to express exposure, service-recency, and cue-effectiveness requirements and score how much a candidate action changes their predicted robustness. This produces an interpretable scalar priority for an existing dispatcher, without requiring joint multi-robot trajectory synthesis.

The application is a decentralized vineyard simulation with reactive detection-response tasks and predictive tasks generated from forecast hotspots. Each robot maintains a local bird-intensity estimate, a cue-effectiveness state reconstructed from available intervention history, and a task queue. No simulator effectiveness state is supplied to the planner. The proposed value compares two local forecasts, with and without a candidate intervention, and scores the difference in specification robustness. Exposure remains the external mission metric, so the evaluation tests whether improving the surrogate translates into improved deterrence.

The contributions are: (i) a counterfactual STL task value that jointly scores exposure, service recency, and cue effectiveness, with an intervention-timed forecast that places suppression at the predicted arrival time; (ii) its integration with cue selection into decentralized reserved-capacity dispatch, using only effectiveness states reconstructed from intervention history; and (iii) an evaluation that isolates the mechanism through a two-factor comparison and a non-habituating control, demonstrates tolerance to depletion-parameter mismatch, and maps the operating envelope across demand and dispatch settings.

A. Related Work and Scope

Multi-robot task allocation provides the broader framework for assigning work under resource and timing constraints [4]. Persistent monitoring regulates fields that grow between robot visits [5]; here, intervention history additionally changes the suppression available at a visit. Time-varying capability can also arise from robot condition. Cline, Macaranas, and Farzan [6] integrate progress-based fault detection with health-aware allocation for heterogeneous teams. Their health estimates modify allocation costs and feasibility masks. Here, the changing state is the effectiveness of an intervention at a location, and the focus is the value assigned to predictive work. This complements health-aware allocation by accounting for changes in the environment’s response to intervention.

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STL describes timed requirements over real-valued signals [7]; quantitative semantics assign satisfaction or violation margins [8], [9]. Smooth approximations [10] and average-based robustness [11] address limitations of minimum-based scoring. We use a soft minimum, not arithmetic-geometric mean semantics. Robustness has also been used in real-time task allocation: Engelaar et al. [12] combine robustness-based filtering, auctions, and probabilistic control synthesis. Liu et al. [13] coordinate assignments and robot progress through STL-constrained optimization. Caballero and Silano [14] use STL for multi-aerial-robot bird-diverter installation. Our distinction is a counterfactual priority for repeated interventions whose effectiveness depletes and recovers, evaluated inside a retained dispatcher. STL supplies the specification and valuation language; execution remains governed by the retained dispatcher.

A self-exciting spatio-temporal point process supplies the bird-risk forecast [15]. The estimator, zone partitioner, and fleet model are supporting infrastructure held fixed across policy comparisons.

II. PROBLEM FORMULATION

A. Domain, Robots, and Task Streams

A vineyard occupies a bounded region $\Omega \subset \mathbb{R}^2$ and is discretized into cells $\mathcal{Z}$. A fleet $\mathcal{R} = \{1, \dots, N\}$ moves through robot zones Z_r , maintained by a health-weighted partitioner. Robot r uses its own cells and a shared boundary band, denoted $\mathcal{Z}_r$, and exchanges boundary information with neighboring robots. The evaluation uses four homogeneous ground robots; the partitioning infrastructure is held fixed.

An action $a = (x_a, c_a, r)$ specifies a target, a cue $c_a \in \mathcal{M}$, and its robot. Its service time $\tau(a, r)$ includes travel and an application hold of $t_{\text{act}} = 20$ s within a 3 m arrival radius. Reactive tasks $\mathcal{A}_r$ originate from confirmed detections and expire T_{react} after arrival. Predictive tasks $\mathcal{A}_p$ originate from forecast hotspots and have no reactive service deadline. Omitting a predictive task incurs no separate deadline penalty but forgoes its possible suppression benefit.

The static importance of location x is

$$w(x) = 1 + w_{\text{row}}(x) + w_{\text{edge}}(x), \quad (1)$$

where the additional terms emphasize row centerlines and field boundaries. The primary outcome is value-weighted bird exposure,

$$J_{\text{exp}} = \int_{t_0}^T \int_{\Omega} w(x) \lambda(x, t) dx dt, \quad (2)$$

where λ is the environment's bird intensity and $[t_0, T]$ is the evaluation window. Reactive deadline misses, response time, and travel are separate operational metrics; they are not explicit penalties in (2).

An empirical load indicator is

$$\rho_{\text{load}} = \frac{\lambda_{\text{react}} \mathbb{E}[\tau]}{N}. \quad (3)$$

where λ_{react} is the reactive-task arrival rate and $\mathbb{E}[\tau]$ the mean service time. Both arrival rate and service time can depend on the policy. Thus ρ_{load} describes the realized

operating regime; it is not an exogenous parameter or a universal stability threshold. We vary background bird intensity to obtain spare, heavy-spare, and high-demand regimes.

B. Effectiveness as a Dynamic State

Each cell–cue pair has effectiveness $\eta_{z,c} \in (0, 1]$. Between applications,

$$\eta_{z,c}(t + \Delta) = 1 - [1 - \eta_{z,c}(t)]e^{-\Delta/T_{\text{rec}}}, \quad (4)$$

and a completed application at time τ produces

$$\eta_{z,c}(\tau^+) = (1 - \kappa)\eta_{z,c}(\tau^-), \quad 0 \leq \kappa < 1. \quad (5)$$

This stylized model represents repetition-induced depletion and recovery during rest. It is not fitted to species-specific field data and does not include transfer of habituation between cue modes. Setting $\kappa = 0$ with unit initial effectiveness removes depletion.

The state summarizes a relevant part of intervention history. For example, if an application leaves $\eta^+ < \eta_{\min} < 1$, the resting time required to recover to the floor is

$$W(\eta^+) = T_{\text{rec}} \log \frac{1 - \eta^+}{1 - \eta_{\min}}. \quad (6)$$

It is zero if $\eta^+ \geq \eta_{\min}$. Since the required rest depends on current depletion, no fixed refractory window can enforce an effectiveness floor; this motivates carrying effectiveness as an explicit state. Cue choice and recovery time then provide two distinct ways to avoid depleted interventions.

C. Bird Intensity and Available Information

The simulated environment combines background intensity, excitation from bird events, and suppression from completed interventions:

$$\lambda(x, t) = [\mu(x)\nu(t) + b(x, t) - u(x, t)]_+, \quad (7)$$

where $\nu(t)$ is a temporal activity profile, with excitation and suppression fields

$$b(x, t) = \sum_{(x_i, t_i) \in \mathcal{E}_t} \alpha K_{\sigma}(x - x_i) e^{-(t-t_i)/\omega}, \quad (8)$$

$$u(x, t) = \sum_{j \in \mathcal{U}_t} q_j \beta_{c_j} K_{\sigma_u}(x - x_j) e^{-(t-\tau_j)/\omega_u}. \quad (9)$$

Here $\mathcal{E}_t$ and $\mathcal{U}_t$ contain past events and completed interventions, K denotes a spatial kernel, α stands for the simulator's self- and cross-excitation weights, β_c is nominal cue strength, and $q_j = \eta_{z_j, c_j}(\tau_j^-)$ is effectiveness immediately before application j depletes the cue. Suppression uses this available effectiveness; the jump in (5) affects later applications. A depleted cue produces less suppression, affecting both exposure and the future reactive workload.

Robot r maintains $\hat{\lambda}_r$ from observed and exchanged events and independently reconstructs cue effectiveness $\hat{\eta}_{r,z,c}$ from the intervention history available locally and from neighboring robots. It propagates this state using (4) and (5) with its own model parameters. The environment maintains a separate effectiveness state $\eta_{z,c}$; the planner neither reads that state nor resets its estimate to it. Counterfactual forecasts operate on copies of the robot's internal state. In the mismatch

study, the planner and environment use different depletion increments, so model error can accumulate in the reconstructed effectiveness as well as affect the candidate forecast.

III. SPECIFICATION-BASED PREDICTIVE TASK VALUE

A. Local Requirements

Robot r monitors, for $z \in \mathcal{Z}_r$,

$$e_z(t) = \int_z w(x) \hat{\lambda}_r(x, t) dx, \quad g_z(t) = t - t_z^{\text{visit}}, \quad (10)$$

together with its reconstructed effectiveness $\hat{\eta}_{r,z,c}$. For brevity, the robot index is suppressed below. The scored specification is

$$\Phi_r^{\text{score}} = \varphi_{\text{exp},r} \wedge \varphi_{\text{cov},r} \wedge \varphi_{\text{hab},r}, \quad (11)$$

where, over a prediction horizon H ,

$$\varphi_{\text{exp},r} = \square_{[0,H]} \bigwedge_{z \in \mathcal{Z}_r} (e_z \leq E^*), \quad (12)$$

$$\varphi_{\text{cov},r} = \square_{[0,H]} \bigwedge_{z \in \mathcal{Z}_r} (g_z \leq T_{\text{cov}}), \quad (13)$$

$$\varphi_{\text{hab},r} = \square_{[0,H]} \bigwedge_{z \in \mathcal{Z}_r} (\text{act}_z \Rightarrow \hat{\eta}_{z,c_z} \geq \eta_{\min}). \quad (14)$$

The last clause applies when a deterrence action is active; c_z is that action's cue. Its purpose is to discourage using an ineffective cue, rather than to require every unused cue to remain above the floor. Inactive cells impose no cue-effectiveness requirement. The same activation convention, normalization, and temporal sampling are used for both counterfactual trajectories.

The normalized predicate margins are

$$m_e = \frac{E^* - e_z}{E^*}, \quad m_g = \frac{T_{\text{cov}} - g_z}{T_{\text{cov}}}, \quad m_\eta = \frac{\hat{\eta}_{z,c_z} - \eta_{\min}}{1 - \eta_{\min}}. \quad (15)$$

The thresholds are positive, with $0 < \eta_{\min} < 1$. Normalization makes the margins dimensionless; it does not remove dependence on the thresholds, which express mission preferences.

Reactive service is monitored separately through

$$\Phi^{\text{react}} = \square \bigwedge_{a \in \mathcal{A}_r} (\text{arr}_a \Rightarrow \Diamond_{[0, T_{\text{react}}]} \text{srva}). \quad (16)$$

This requirement is handled by the dispatcher's deadline-priority rule (Table I) rather than by (11), and its satisfaction is reported as a separate metric.

B. Robustness Aggregation

Standard STL robustness evaluates a predicate by its margin, conjunction by a minimum, and bounded always by an infimum over time [8], [9]. Positive robustness implies satisfaction and negative robustness implies violation; the zero boundary requires separate treatment. On the sampled forecast, temporal extrema are evaluated over monitor samples separated by Δ_m .

Let R_i denote the normalized robustness values being conjoined. We use the log-sum-exp soft minimum [10],

$$\tilde{R} = \text{smin}_\theta(R_1, \dots, R_k) = -\frac{1}{\theta} \log \sum_{i=1}^k e^{-\theta R_i}, \quad \theta > 0. \quad (17)$$

Its elementary properties explain its role:

$$\min_i R_i - \frac{\log k}{\theta} \leq \tilde{R} \leq \min_i R_i, \quad \frac{\partial \tilde{R}}{\partial R_i} = \frac{e^{-\theta R_i}}{\sum_j e^{-\theta R_j}}. \quad (18)$$

Thus a conjunctive aggregate can respond to improvements in several clauses instead of only its current minimum. The surrogate places greater weight on smaller margins while retaining sensitivity to other conjoined values. Hard temporal or spatial minima can still mask changes in non-binding samples. The method ranks discrete candidates and does not require gradient-based trajectory optimization.

C. Counterfactual Action Value

For candidate a , let ξ^{-a} denote the local forecast without that candidate and ξ^{+a} the forecast with its intervention. The two forecasts begin from the same available state and differ by the candidate's modeled consequences. The predictive task value is

$$U(a, r) = \tilde{R}(\Phi_r^{\text{score}}, \xi^{+a}, t) - \tilde{R}(\Phi_r^{\text{score}}, \xi^{-a}, t). \quad (19)$$

For a target location, cue selection maximizes U over available modes. This need not coincide with choosing the freshest cue when modes differ in suppression strength or execution cost. The greedy habituation-aware comparator instead uses the freshest cue at its selected target.

D. Intervention-Timed Forecast

Let $t_a = t + \delta$ be the candidate's predicted intervention time. Both forecasts share the same background assumptions and initial event history. Existing excitation and inhibition are advanced by passive decay; for forecast offset $s \geq 0$,

$$\hat{\lambda}_r^-(x, t + s) \approx [\mu(x)\nu(t + s) + \text{trig}_r(x, t)e^{-s/\omega} - \text{inh}_r(x, t)e^{-s/\omega_u}]_+. \quad (20)$$

This approximation does not integrate all future offspring events. It also simplifies the environment model in two respects: past suppression is advanced with the nominal cue strength β_c rather than the effectiveness-scaled $q_j\beta_{c_j}$, and excitation is summarized by a single kernel weight rather than the simulator's separate self- and cross-excitation terms. Both simplifications enter ξ^{-a} and ξ^{+a} identically, so they affect the absolute forecast rather than the difference that defines $U(a, r)$. The candidate's effectiveness is taken from the reconstructed state at planning time,

$$\hat{q}_a = \hat{\eta}_{z_a, c_a}(t), \quad (21)$$

without advancing recovery over the travel interval δ . Since recovery can only raise $\hat{\eta}$ and δ is bounded by the predictive lead settings, this is a conservative approximation that slightly understates the suppression of a candidate whose cue is still recovering. The post-application state is $\hat{\eta}_{z_a, c_a}(t_a^+) = (1 - \hat{\kappa})\hat{q}_a$, where $\hat{\kappa}$ is the planner's assumed depletion increment. The added suppression field is

$$h_a(x, t + s) = \begin{cases} 0, & s < \delta, \\ \hat{q}_a \beta_{c_a} K_{\sigma_u}(x - x_a) e^{-(s-\delta)/\omega_u}, & s \geq \delta. \end{cases} \quad (22)$$

Thus suppression starts at the intervention and decays from that instant, not from the planning time. Figure 1 illustrates

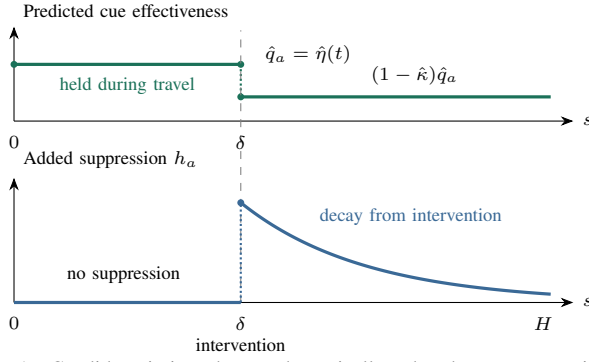


Fig. 1. Candidate timing, shown schematically rather than as an experimental trace. During travel, candidate suppression is zero and the cue estimate is held at its planning-time value. At $t_a = t + \delta$, suppression uses the pre-depletion effectiveness $\hat{q}_a$ and subsequently decays; the depletion jump affects later applications.

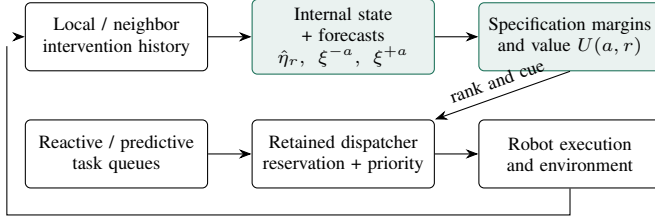


Fig. 2. Per-robot valuation and dispatch. The robot reconstructs cue effectiveness from locally available and exchanged intervention history. Both counterfactuals use copies of this internal state, with no simulator-state access. Shaded blocks define predictive priority; urgency remains a separate decision.

these two time scales. The corresponding intensity reduction is capped by the baseline forecast,

$$d_a(x, t + s) = \min\{\hat{\lambda}_r^-(x, t + s), h_a(x, t + s)\}, \quad (23)$$

and the exposure signal in every affected cell is updated as

$$e_z^{+a}(t + s) = e_z^{-a}(t + s) - \int_z w(x) d_a(x, t + s) dx. \quad (24)$$

Completion also resets service recency and applies the modeled depletion update. The comparison consequently captures the spatial extent and timing of suppression and the change in the effectiveness state, while keeping the background forecast identical.

E. Integration with Dispatch

Figure 2 summarizes the per-robot pipeline. Each robot tracks the fraction χ_r of a sliding window T_W spent on predictive work. Table I retains the reservation ρ_{res} and deadline-priority rule. The new value changes predictive ranking and cue selection, while task admission and the dispatch structure remain fixed among the predictive systems. The comparison with greedy dispatch is consequently a system-level comparison. The retained dispatcher ranks admitted predictive tasks without an additional positivity gate; its argmax need not be positive. Global exposure is evaluated independently because improving local robustness is a surrogate for the mission objective.

IV. EVALUATION DESIGN

A. Simulation and Comparators

Table II reports production settings recorded in the per-run configuration exports. All systems use the same bird-

TABLE I

DISPATCH PRIORITY AT EACH DECISION. APPLY THE FIRST ELIGIBLE RULE.

Order	Condition	Selected task
1	Earliest reactive expiration is within τ_{exp}	That reactive task
2	$\chi_r < \rho_{\text{res}}$ and a predictive task is admitted	Predictive task with maximum $U(a, r)$
3	A reactive task is available	Highest-value reactive task
4	A predictive task is admitted	Predictive task with maximum $U(a, r)$
5	Otherwise	Idle

TABLE II

RECORDED PRODUCTION SETTINGS. TIMES ARE IN SEC. UNLESS NOTED.

Setting	Value
Field; grid; fleet	500 × 500 m; 60 × 48; 4 UGVs
Run duration; warm-up	1800; none
Simulation step; replanning	1; 45
Arrival radius; action hold	3 m; $t_{\text{act}} = 20$
Reservation; accounting window	$\rho_{\text{res}} = 0.25$; $T_W = 600$
Deadline-priority slack	$\tau_{\text{exp}} = 90$ (sweep: 30–90)
Predictive lead settings	Minimum 30; maximum ETA 120; buffer 15
Depletion	Sweep: 0–0.80; mismatch: 0.10–0.90; confirmatory: 0.50

process family, fleet, spatial weights, and supporting estimation infrastructure. Specification thresholds, prediction horizon, smoothing parameter, and cue parameters are held fixed across the principal comparisons; the component comparisons change only their stated factors.

Table III defines the five main systems. B2 retains the existing counterfactual exposure-reduction value ΔJ . B3 uses exposure and coverage robustness with a fixed cue. B4 combines the full scored specification with flexible cue selection. B5 keeps greedy task dispatch and selects the freshest cue. The B4–B5 contrast tests the practical value of the complete predictive system beyond cue choice alone. The two-factor comparison in Sec. V-B holds predictive dispatch fixed to examine the interaction of specification and cue flexibility.

The configured background rates are $\mu \in \{2 \times 10^{-5}, 10^{-4}, 4 \times 10^{-4}\}$ per cell per second. For the spatial-density notation in (7), the corresponding rates are divided by cell area. We call these spare, heavy-spare, and high-demand conditions throughout; the figure legends label the last “overloaded.” Across the joint sweep, observed load indicators are approximately 0.39–0.52, 0.50–0.86, and at least 0.88, respectively. Habituation strength is $\kappa \in \{0, 0.10, 0.25, 0.40, 0.50, 0.65, 0.80\}$. A non-habituating control fixes environment effectiveness to one. The mismatch study fixes planner $\kappa = 0.50$ and varies environment κ .

B. Outcome Summaries and Statistical Interpretation

Systems are paired by random seed with randomness controlled consistently across policies. Bootstrap resampling preserves complete seed pairs. The joint load-habituation sweep uses 10 seeds per cell. The spare-load baseline comparison, two-factor comparison, and dedicated B4–B5 comparisons use 30; the mismatch and override-slack studies use 10, and the preemption study uses 30. Each summary is tied to its configuration, seed set, and analysis script. Studies

TABLE III

MAIN COMPARATORS. PREDICTIVE SYSTEMS SHARE THE DISPATCHER.

ID	Dispatch	Predictive value	Cue choice
B1	Greedy	None	Fixed
B2	Reserved	Exposure ΔJ	Fixed
B3	Reserved	STL exposure + coverage	Fixed
B4	Reserved	Full Φ_r^{score}	Maximize U
B5	Greedy	None	Freshest

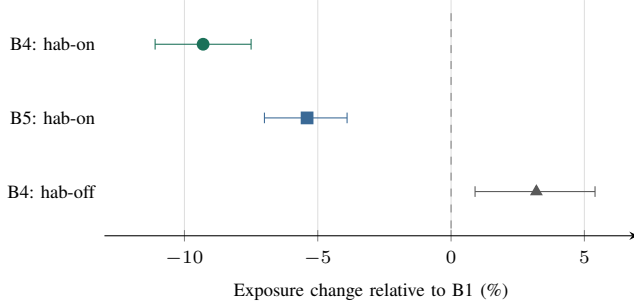


Fig. 3. Spare-load comparison ($\mu = 2 \times 10^{-5}$, $\kappa = 0.50$, 30 paired seeds). Points and intervals show paired exposure effects and bootstrap 95% intervals. The benefit is specific to habituation: the non-habituating control shows the cost of the method when there is no depletion to exploit.

with different sample sizes or summary definitions are kept separate rather than pooled.

Exposure effects are expressed relative to the corresponding B1 reference; negative values favor the evaluated policy. A B4–B5 difference expressed as a percentage of B1 is not a percentage reduction relative to B5. Confidence intervals are paired bootstrap 95% intervals. Where listed, p -values are two-sided Wilcoxon signed-rank values and are nominal unless explicitly described as Holm-adjusted. The exploratory grid uses Holm correction. Reanalysis of the existing paired runs preserves the B4–B5 improvement under multiplicity control; the table retains nominal p -values. Effect sizes and intervals are the primary summaries; nonsignificant comparisons are not interpreted as equivalence or noninferiority.

Reactive miss rate is the fraction of detection-triggered tasks expiring before service. Mean response time, travel, coverage robustness, effectiveness at application, and cell-level cue-use entropy provide operational and mechanism diagnostics. Because deterrence changes future detections, both the number and composition of reactive tasks can change across policies; miss-rate comparisons are therefore accompanied by exposure and, where available, task counts.

V. RESULTS

A. Exposure Benefit Beyond Greedy Cue Choice

At spare load with $\kappa = 0.50$, B4 reduces exposure by 9.3% of B1, with a paired 95% interval of $[-11.1, -7.5]\%$. Greedy freshest-cue selection (B5) yields a 5.4% reduction, with interval $[-7.0, -3.9]\%$. Prediction alone does not produce this gain: the legacy exposure predictor (B2, +0.4%) and STL without the effectiveness clause or cue flexibility (B3, -1.1%) have intervals spanning zero. The improvement is therefore tied to history-dependent valuation with cue selection. Figure 3 shows the two effective systems and the non-habituating control.

TABLE IV

DEDICATED B4 VERSUS B5 COMPARISON ($\kappa = 0.50$, HAB-ON). DIFFERENCES ARE PERCENTAGES OF B1; p -VALUES ARE NOMINAL.

Demand	n	B4–B5	p	B4 wins
Spare	30	-3.9%	0.002	23/30
Heavy-spare	30	-3.1%	0.006	22/30
High-demand	30	$+4.8\%$	< 0.001	3/30

TABLE V

TWO-FACTOR COMPARISON AT SPARE LOAD, HAB-ON, $\kappa = 0.50$, $n = 30$. ALL CELLS USE RESERVED DISPATCH. ENTRIES ARE EXPOSURE CHANGES RELATIVE TO B1. THE INTERACTION IS DESCRIPTIVE.

Scored specification	Fixed cue	Flexible cues
Exposure + coverage	-1.1%	$+1.1\%$
Full specification	$+1.1\%$	-9.3%

The dedicated paired comparison in Table IV shows an additional benefit of B4 over B5 in both spare-capacity regimes: 3.9 and 3.1 percentage points relative to B1. At spare load, the paired absolute difference has a 95% interval of $[-2506, -730]$ exposure units, and B4 performs better in 23 of 30 pairs. The proposed system thus improves on a strong habituation-aware baseline, not only on a fixed-cue reference. Section V-B holds dispatch fixed to isolate the roles of the effectiveness clause and cue flexibility.

B. Effectiveness-Aware Valuation and Cue Flexibility

Table V crosses inclusion of the effectiveness clause with cue flexibility. Reserved dispatch and all other configuration choices are held fixed, so the four systems differ only in these two factors. The joint configuration achieves -9.3% , whereas either isolated change produces $+1.1\%$ in this comparison. The joint configuration therefore improves on each single-component variant by 10.4 percentage points relative to B1. The two components are complementary: the effectiveness requirement changes which interventions are prioritized, and cue flexibility lets the robot act on that information. Neither suffices alone, and paired analysis confirms that the full method outperforms both single-component variants.

C. Dependence on Habituation and Demand

The non-habituating control acts as a falsification test. With habituation disabled, B5 is approximately equal to B1 and B3 is at $+0.1\%$, while B4 incurs a $+3.2\%$ spare-load exposure cost, with interval $[+0.9, +5.4]\%$. The advantage of B4 thus arises from exploiting cue depletion, and the control quantifies the cost of deploying it when habituation is absent.

Figure 4 shows the joint load–habituation sweep. With no depletion, B4 does not improve exposure. Once depletion is nonzero, the two spare-capacity curves move into the beneficial region, with a magnitude that varies with κ . The high-demand curve remains above the greedy reference. The benefit is therefore habituation-dependent and conditional on spare service capacity.

The Holm-adjusted exploratory comparisons include improvements at spare and heavy-spare load and reversals at

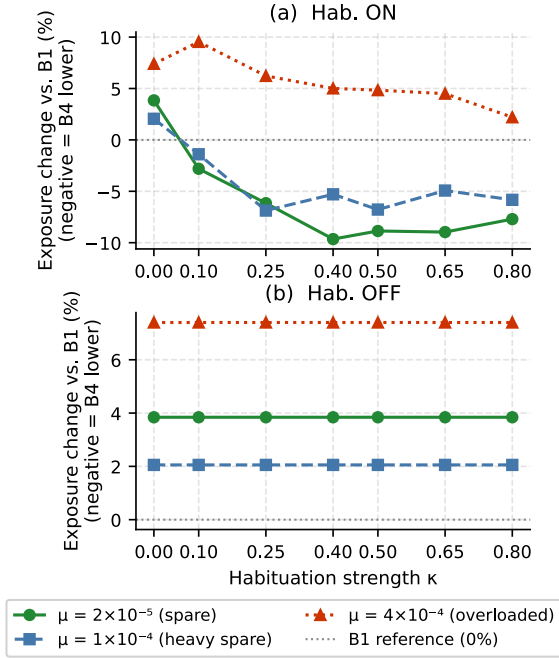


Fig. 4. Joint sweep ($n = 10$ per cell): B4 exposure change relative to B1 under habituating and non-habituating conditions. Curves show study summaries, not confidence bands. Benefits emerge in the spare-capacity regimes but are not monotonic in κ ; the high-demand regime retains a penalty.

high demand, consistent with Table IV. In the tested scenarios the reversal is bracketed by observed load indicators of 0.86 and 0.88, which suggests ρ_{load} as a practical signal for load-aware supervision.

Mechanism diagnostics are consistent with this account. At $\kappa = 0$ the sweep gives approximately unit effectiveness and negligible cue-use entropy; as depletion increases, cue variety increases in the spare-capacity regimes. Under high demand, effectiveness at application falls to approximately 0.14–0.18 and cue variety becomes small: when all cues are depleted, there is little for effectiveness-aware valuation to select among.

D. Robustness to Depletion-Parameter Mismatch

The benefit does not require knowing the depletion parameter. Table VI fixes planner $\kappa = 0.50$ and varies environment κ from 0.10 to 0.90 at heavy-spare load. B4 has lower mean exposure than B1 at all five levels (nominal $p \leq 0.008$ at four; $p = 0.053$ at the weakest depletion) and improves over B3 throughout. Because the planner propagates its own effectiveness state with the assumed κ , these results include the accumulated state-model discrepancy, not only a one-step forecast error.

E. Exposure–Service Trade-off

B4 reallocates effort from individual detections to field-level suppression, and the cost appears in reactive service. Table VII shows B4 miss-rate increases of 17.7, 20.4, and 13.4 percentage points relative to B1, with mean response time increasing by 19.2, 24.3, and 31.1 s. Travel increases substantially at heavy-spare and high demand. B3 incurs a comparable penalty, so most of the trade-off is a property of

TABLE VI
DEPLETION-PARAMETER MISMATCH AT HEAVY-SPARE LOAD ($n = 10$).
PLANNER $\kappa = 0.50$. DIFFERENCES ARE EXPOSURE UNITS; LISTED
 p -VALUES ARE NOMINAL.

Environment κ	B4–B1	p	B4–B3	p
0.10	−9489	0.053	−13451	0.006
0.25	−12765	0.006	−19515	0.006
0.50	−13648	0.008	−19386	0.006
0.75	−14327	0.006	−16611	0.006
0.90	−13110	0.006	−16323	0.006

TABLE VII
OPERATIONAL METRICS AT $\kappa = 0.50$, HAB-ON ($n = 10$ PER
CONDITION). RT IS MEAN REACTIVE RESPONSE TIME; TRAVEL IS THE
B4/B1 FLEET-DISTANCE RATIO.

Demand	Miss rate (%)			RT (s)		Travel B4/B1
	B1	B3	B4	B1	B4	
Spare	24.7	35.4	42.4	80.6	99.8	1.06
Heavy-spare	26.7	46.4	47.1	74.8	99.1	1.45
High-demand	52.4	66.3	65.8	70.2	101.3	1.46

predictive dispatch; the spare-load B4 miss rate exceeds B3’s, indicating an additional contribution from the valuation.

In the separate 30-seed spare-load comparison, B1 generates 32.7 reactive tasks and completes 26.3 per run, while B4 generates 29.9 and completes 18.5. B4 therefore lowers both exposure and the number of detections that require a response, while completing fewer of the responses that remain. Coverage robustness does not differ significantly between B1 and B4.

F. Persistence Across Dispatch Settings

We test whether the trade-off is an artifact of dispatcher tuning. Figure 5 allows urgent reactive work to preempt predictive assignments: the miss-rate gap narrows modestly in some conditions but remains positive, and the high-demand exposure penalty remains. Figure 6 compares priority slacks of 30, 45, 60, and 90 s and travel-cap variants; the 45 s setting has favorable means at heavy-spare load, though its intervals overlap the default. A reservation sweep at high demand, including $\rho_{\text{res}} = 0$, also retains a positive B4–B1 exposure gap, because predictive generation, ranking, and idle-capacity use remain active.

Across all tested modifications the pattern is the same: spare-capacity exposure benefits with increased reactive misses, and a high-demand penalty. The trade-off thus reflects the allocation of effort between predictive and reactive work, not a particular dispatcher setting.

VI. DISCUSSION

The results give a consistent account. History-dependent valuation is effective when two conditions hold: cue effectiveness depletes, and the fleet has capacity to act on that information. The two-factor comparison shows that the effectiveness requirement and cue flexibility are complementary, the non-habituating control shows that the gain comes from exploiting depletion, and the mismatch study shows that the

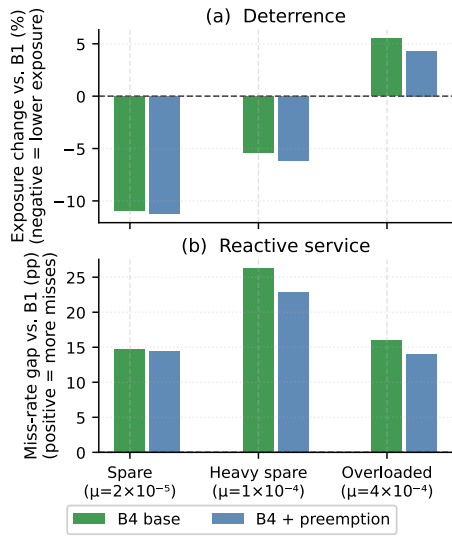


Fig. 5. Dedicated preemption experiment ($n = 30$, hab-on, $\kappa = 0.50$). Mean exposure and miss-rate changes relative to B1 are shown for the base and preemptive variants. Only means are displayed; the changes are interpreted descriptively.

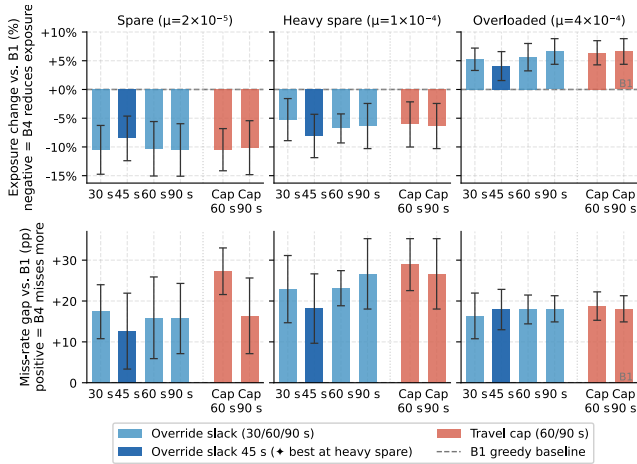


Fig. 6. Dispatch-sensitivity experiment ($n = 10$): exposure changes and miss-rate gaps relative to B1 for priority-slack and travel-cap variants. Error bars show 95% confidence intervals.

gain survives large error in the assumed depletion increment. Normalized specification margins make the competing preferences explicit: suppressing risk, revisiting neglected locations, and avoiding depleted cues. The same construction applies to other repeated interventions whose effectiveness depletes and recovers.

The evaluation also delineates where the method should not be used. Under sustained high demand, habituation-aware greedy dispatch (B5) is the better tested option, and under spare capacity the exposure gain is traded against reactive deadline misses. Both observations point to the same extension: load-aware supervision that switches policies using ρ_{load} , and incorporating reactive service into predictive valuation or admission.

VII. CONCLUSION

We developed a counterfactual STL-based task value for repeated robotic interventions with depleting and recovering effectiveness. With spare capacity, combining an

effectiveness requirement with flexible cue selection reduces exposure by 9.3% relative to fixed-cue greedy dispatch and by a further 3.9 percentage points beyond habituation-aware greedy dispatch, and the gain is robust to mismatch in the depletion parameter. Component comparisons and controls attribute the gain to the joint use of effectiveness-aware valuation and cue choice, and map its operating envelope: the benefit requires active habituation and spare capacity, and is traded against reactive service in a way that persists across dispatch modifications. Future work will add load-aware supervision and reactive-service terms to the valuation, and calibrate the stylized habituation model against field data.

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